

SIH26181 — Master Theory Notes & Judge Defense Companion

AI-Powered Personal Health Companion & Edge Disaster Monitor

FPGA-Accelerated, Cloud-Free Biometric & Disaster Resilience Engine

SIH26181 — Master Theory Notes & Judge Defense Companion

AI-Powered Personal Health Companion & Edge Disaster Monitor

Qualcomm Hardware Challenge — Smart India Hackathon 2026

Target Audience: 2nd-Year Engineering Students & Team Members

Goal: Plain-English, comprehensive guide covering **what our project is, why it matters, how every piece works, all technical terms from A to Z, and exact answers to tough questions judges will ask.**

Table of Contents

- Part 0: The Big Picture (Start Here!)
- 0.1 The 60-Second Elevator Pitch
- 0.2 The Real-World Problem in India
- 0.3 The Solution: Heterogeneous Edge Computing (FPGA + CPU)
- 0.4 Complete Hardware & Software Bill of Materials (A to Z)
- Part 1: What We Are Actually Doing — End-to-End Pipeline
- 1.1 The Complete 6-Step Data Flow
- 1.2 Hardware vs. Software Division of Labor
- 1.3 Memory-Mapped Registers in Action (0x00 – 0x14)
- 1.4 The 3 Real-World Disaster Scenarios
- Part 2: Complete Concept & Theory Glossary (55+ Terms)
- 2.1 Sensors & Medical Science
- 2.2 Signal Processing & DSP
- 2.3 FPGA & Digital Design
- 2.4 Bus Protocols & Communication
- 2.5 Verification & Timing
- 2.6 Software & Embedded Systems
- 2.7 Disaster Science & Risk Indices

- Part 3: Edge AI & TinyML Neural Network Engine
- 3.1 Why Machine Learning on Top of Formulas?
- 3.2 The 6→12→3 Network Architecture
- 3.3 How Inference Works in Under 1 Microsecond
- Part 4: Qualcomm Platform Strategy & Silicon Migration
- 4.1 Why Prototype on FPGA and Deploy on Qualcomm Silicon?
- 4.2 Qualcomm Snapdragon Wear W5+ Gen 1 & QCS6490 Mapping
- 4.3 Qualcomm Hexagon DSP & Low-Power Island (LPI)
- 4.4 Qualcomm AI Engine (SNPE / QNN)
- Part 5: The Ultimate Judge Defense Guide (Top 25 Q&A)
- 5.1 Hardware & FPGA RTL Questions
- 5.2 Signal Processing & Biomedical Questions
- 5.3 Embedded Systems & Protocol Questions
- 5.4 AI & Qualcomm Strategy Questions
- Part 6: 60-Second Presentation Pitch for Students

Part 0: The Big Picture (Start Here!)

0.1 The 60-Second Elevator Pitch

*"We built an **FPGA-accelerated personal health companion and edge disaster monitor**. During extreme heat waves, severe air pollution, or floods, people suffer organ damage or death because physiological warning signs (like heart rate variability collapse and blood oxygen drop) happen invisibly inside the body. Our device pairs **custom Verilog hardware** that filters noisy optical signals and extracts heartbeat timing with **20-nanosecond precision at zero CPU load**, with an on-device **TinyML neural network** and disaster risk engine. It fuses the user's vitals with environmental temperature, humidity, and PM2.5 levels to predict heat exhaustion and respiratory distress **15 to 30 minutes before clinical symptoms occur** — completely offline, with **zero cloud or internet dependence**."*

0.2 The Real-World Problem in India

India faces severe recurring environmental emergencies:

- 1. Extreme Heat Waves:** Temperatures regularly exceed 45°C–48°C in North and Central India. High humidity prevents sweat from evaporating. The heart is forced to pump faster and faster while blood volume drops (*Cardiovascular Drift*). By the time someone faints from heat stroke, internal organ damage has already started.
- 2. Winter Air Pollution Smog:** Delhi/NCR PM2.5 levels frequently cross 400 µg/m³ (hazardous). Toxic particulates block oxygen transfer in the lungs, forcing the heart into compensatory tachycardia while blood oxygen (SpO₂) plummets.
- 3. Disaster Zone Isolation:** In floods, earthquakes, or industrial fires, cellular towers and power grids go down. **Cloud-connected smartwatches (Apple Watch, Fitbit) become useless paperweights because they rely on cloud servers to run AI analytics.**

0.3 The Solution: Heterogeneous Edge Computing (FPGA + CPU)

We use a **System-on-Chip (SoC)** architecture that splits work intelligently between hardware and software:

HETEROGENEOUS SYSTEM-ON-CHIP (SoC)	
PROGRAMMABLE LOGIC (FPGA RTL) "Fast, Jitter-Free Reflexes"	PROCESSING SYSTEM (ARM CPU) "Complex High-Level Brain"
• 50 MHz Custom Hardware Engine • Dual-channel 8-tap DSP Filter • 4-State Peak Detector FSM • 20 ns Cycle-Accurate IBI Timer • Zero CPU Load & Zero OS Jitter • Standard ARM AMBA AXI4-Lite Bus	• Bare-Metal C Runtime / Drivers • HRV Mathematical Analysis (RMSSD) • SpO₂ Beer-Lambert Calibration • TinyML 6→12→3 Neural Network • Multi-Disaster Risk Score Engine • SSD1306 OLED User Interface

Why not do everything in software on an Arduino or Raspberry Pi?

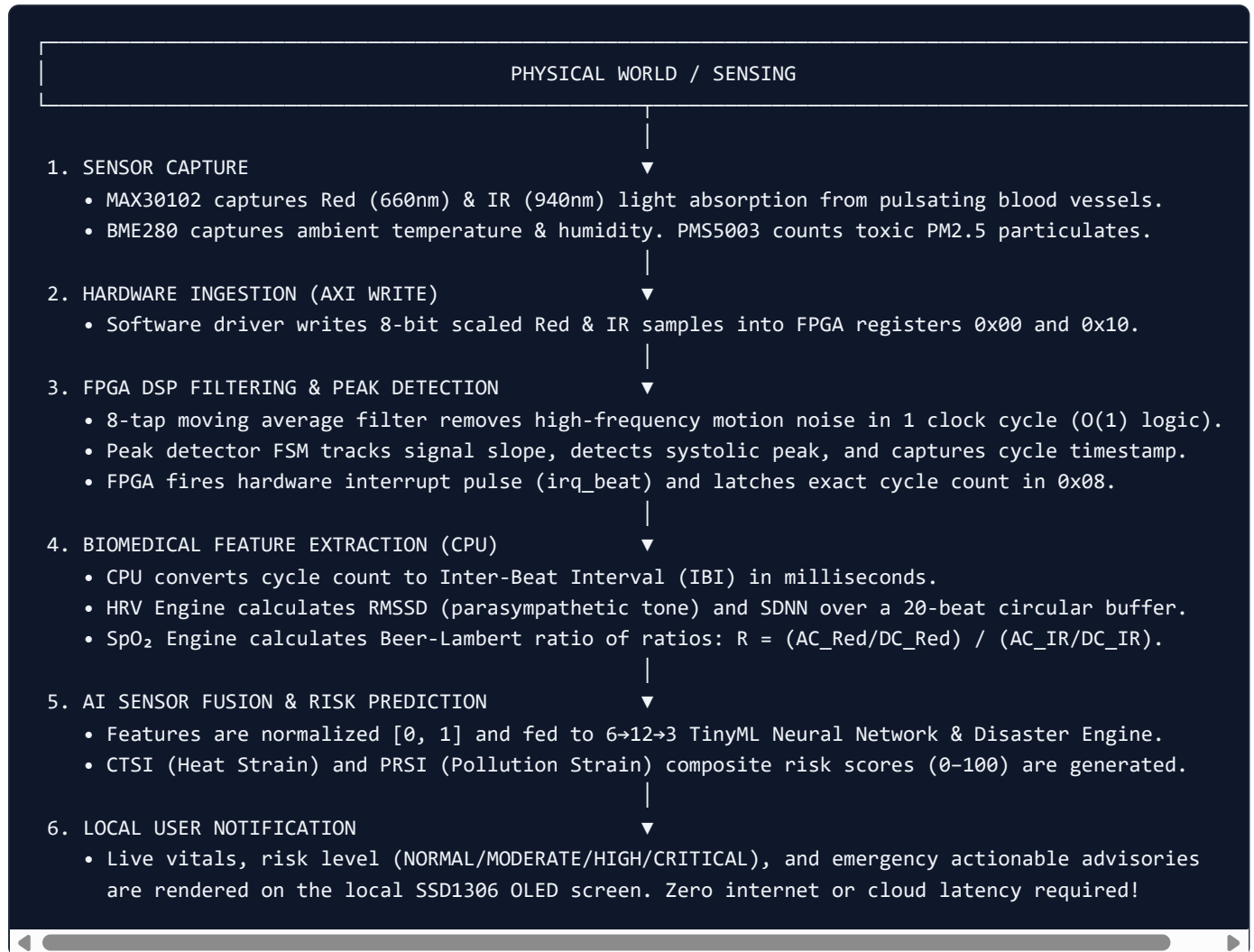
- **Software Jitter:** An operating system (like Linux or RTOS) or a software microcontroller has interrupt latency and thread scheduling delays of **5 to 20 ms**. If your heartbeat interval is 800 ms, a 20 ms jitter is a massive **2.5% error**, which completely ruins Heart Rate Variability (HRV) calculations.
- **FPGA Hardware Precision:** Our FPGA runs a dedicated 32-bit hardware timer at **50 MHz**, measuring each heartbeat interval with **20 nanoseconds resolution (250,000× more precise)** with **0 CPU overhead**.

0.4 Complete Hardware & Software Bill of Materials (A to Z)

Component	Category	Purpose in Project	Interface / Protocol
MAX30102	Sensor	Dual-LED (660nm Red + 940nm IR) optical pulse oximeter & heart rate sensor	I ² C (Address <code>0x57</code>), 18-bit ADC
BME280	Sensor	Environmental temperature, relative humidity, barometric pressure	I ² C (Address <code>0x76</code>), Bosch math
PMS5003	Sensor	Laser particulate matter sensor (measures PM1.0, PM2.5, PM10)	UART (9600 baud, 32-byte frames)
SSD1306	Display	0.96-inch monochrome OLED (128×64 pixels) for real-time vitals/advisories	I ² C (Address <code>0x3C</code>), 1024-byte buffer
Xilinx Zynq-7000 (<code>xc7z020</code>)	Target SoC (Prototype)	Dual-core ARM Cortex-A9 CPU + Artix-7 FPGA fabric on single silicon	AMBA AXI4-Lite Memory-Mapped Bus
Qualcomm Snapdragon Wear W5+ Gen 1	Target SoC (Production)	Commercial low-power wearable AP + Hexagon DSP + NPU	Qualcomm System NoC + QUP I ² C/UART
<code>axippgaccelerator.v</code>	Verilog RTL	Top-level hardware IP with 6 memory-mapped registers & decoupled AXI slave	AXI4-Lite Slave
<code>movingaverage8tap.v</code>	Verilog RTL	O(1) running-sum 8-tap low-pass noise filter (0 DSP slices, 0 BRAM)	Pure Combinational/Sequential RTL
<code>ppgpeakdetector.v</code>	Verilog RTL	4-state FSM with 250 ms refractory blanking & 20 ns cycle counter	Hardware Interrupt (<code>irq_beat</code>)
<code>nnriskmodel.c</code>	Embedded C	On-device TinyML Neural Network (6→12→3 architecture, 108 MACs)	Header API (<code>nnpredictrisk</code>)
<code>disasterriskengine.c</code>	Embedded C	Rule-based multi-disaster engine (CTSI, PRSI, Hypothermia scoring)	Header API (<code>disasterassessrisk</code>)
<code>tbppgsystem.v</code>	Verification	6-test self-checking simulation testbench with Bus Functional Model (BFM)	Icarus Verilog (<code>iverilog</code> / <code>vvp</code>)
GTKWave / <code>signals.gtkw</code>	Tool	Visual waveform analysis tool for inspecting signal transitions	VCD Waveform Dump (<code>ppg_system.vcd</code>)
Vivado ML Standard	Tool	Synthesis, Placement, Routing, and Static Timing Analysis (STA)	TCL Batch Script (<code>runvivadosynth.tcl</code>)

Part 1: What We Are Actually Doing — End-to-End Pipeline

1.1 The Complete 6-Step Data Flow



1.2 Hardware vs. Software Division of Labor

A common judge question is: "Why did you put this part in Verilog and that part in C?"

Task	Where It Runs	Technical Rationale
Noise Filtering (Moving Average)	FPGA (Verilog)	Needs continuous, streaming execution with zero CPU load. Implemented as O(1) hardware shift-registers with 1-cycle throughput.
Peak Detection & IBI Timing	FPGA (Verilog)	Requires microsecond/nanosecond accuracy (20 ns) to eliminate software OS scheduling jitter.
Sensor Register Config & I2C/UART	CPU (C Driver)	Low-speed protocol configuration is simpler and more flexible in C software.
HRV (RMSSD) & SpO ₂ Mathematics	CPU (C Math)	Square roots, floating-point calibration, and division are cheaper on an ARM CPU with an FPU than wasting FPGA DSP slices.
Multi-Sensor Fusion & TinyML AI	CPU / Hexagon NPU	Neural network matrix multiplication (6→12→3) executes in under 1 μs in C or on Qualcomm AI Engine.
OLED Display Rendering	CPU (C HAL)	Font rendering, screen layout, and string formatting belong in software.

1.3 Memory-Mapped Registers in Action (0x00 – 0x14)

Our FPGA accelerator acts as a slave on the ARM AMBA AXI4-Lite bus mapped at address 0x43C00000 :

Offset	Register Name	Access	Reset Value	Function & Bit Fields
0x00	REG_RED_RAW	R/W	0x00000000	[7:0] Writing raw Red sample triggers DSP filter
0x04	REG_RED_FILTERED	RO	0x00000000	[7:0] Smoothed Red output (read by SpO ₂ engine)
0x08	REG_IBI_CYCLES	RO	0x00000000	[31:0] Inter-Beat Interval count in 20ns clock tick
0x0C	REG_STATUS_THRESH	Mixed	0x00007800	[0] beat_flag (W1C: Write-1-to-Clear) [15:8] Systolic detection threshold (Default: 120)
0x10	REG_IR_RAW	R/W	0x00000000	[7:0] Writing raw IR sample triggers DSP filter
0x14	REG_IR_FILTERED	RO	0x00000000	[7:0] Smoothed IR output (read by SpO ₂ engine)

1.4 The 3 Real-World Disaster Scenarios

Our software simulation (mainsimulation.c / health demo.exe) tests three distinct disaster events:

Scenario 1: Normal Baseline Condition

- **Environment:** 25°C, 45% Humidity, PM2.5 = 15 μg/m³ (Clean Air).
- **Vitals:** Heart Rate = 72 BPM, SpO₂ = 98%, RMSSD = 45 ms (Healthy parasympathetic tone).
- **System Output:** **NORMAL RISK (Score: 12/100)** → "Vitals stable. All physiological parameters within healthy baseline."

Scenario 2: Delhi Extreme Heat Wave (47°C, 65% Humidity)

- **What happens to the body:** Body blood vessels dilate to radiate heat. Blood pools in extremities, reducing venous return to the heart. To maintain blood pressure, the heart accelerates (**Cardiovascular Drift**), and the autonomic nervous system is pushed to exhaustion.

- **Vitals Evolution:** Heart Rate climbs from 72 → 140 BPM, while RMSSD collapses from 45 → 8 ms.
- **System Output: CRITICAL HEAT STRAIN (CTSI Score: 78/100)** → *"DANGER: Heat stroke imminent! Cardiovascular drift detected. Cease physical exertion, seek active cooling immediately."*
- **Clinical Value:** Gives **15 to 30 minutes of advance warning** before the person passes out or suffers organ damage.

Scenario 3: Delhi Winter Smog (PM2.5 = 400 µg/m³, Severe Inversion)

- **What happens to the body:** Fine particulate matter (< 2.5 µm) fills lung alveoli, impairing oxygen diffusion into the bloodstream. Blood oxygen drops, and the heart speeds up to compensate (*Compensatory Tachycardia*).
- **Vitals Evolution:** SpO₂ drops from 98% → 86%, Heart Rate rises to 115 BPM, RMSSD = 16 ms.
- **System Output: CRITICAL RESPIRATORY DISTRESS (PRSI Score: 85/100)** → *"DANGER: Severe hypoxemia from particulate inhalation. Move to filtered shelter, administer supplemental oxygen."*

Part 2: Complete Concept & Theory Glossary (55+ Terms)

2.1 Sensors & Medical Science

1. PPG (Photoplethysmography)

- **What it is:** An optical method to detect blood volume changes in microvascular tissue using light.
- **How it works:** An LED illuminates tissue; when the heart contracts (systole), more blood fills the capillaries, absorbing more light. When the heart rests (diastole), less blood is present, absorbing less light. A photodetector measures this light variation, producing a pulsatile wave.
- **Analogy:** Like holding a flashlight against your finger and watching the light glow pulse with your heartbeat.

2. Dual-Wavelength Pulse Oximetry (660nm vs 940nm)

- **Why two lights?** Oxygenated hemoglobin (HbO₂) and deoxygenated hemoglobin (Hb) have drastically different light absorption spectra:
- **Red Light (660 nm):** Deoxygenated blood absorbs *more* Red light.
- **Infrared Light (940 nm):** Oxygenated blood absorbs *more* Infrared light.
- **Formula:** By calculating the Ratio of Ratios $R = (AC_{Red} / DC_{Red}) / (AC_{IR} / DC_{IR})$, we determine oxygen saturation.

3. SpO₂ (Blood Oxygen Saturation)

- **What it is:** Percentage of hemoglobin molecules in arterial blood that are saturated with oxygen.
- **Clinical Scale:**
 - 95% - 100%: Normal healthy individual.
 - 90% - 94%: Mild Hypoxemia (shortness of breath, fatigue).
 - < 90%: Critical Medical Emergency (potential organ hypoxia).
- **Beer-Lambert Calibration in our code:** $SpO_2 = 110 - 25 \times R$.

4. IBI (Inter-Beat Interval)

- **What it is:** The precise time in milliseconds between two consecutive heartbeat peaks (R-R interval).
- **Formula:** $IBI \text{ (ms)} = 60,000 / \text{Heart Rate (BPM)}$. For 75 BPM, $IBI = 800 \text{ ms}$.

5. HRV (Heart Rate Variability)

- **What it is:** The subtle, natural variation in time intervals between consecutive heartbeats.
- **Why it matters:** A healthy heart is **not** a rigid clock metronome — it continuously speeds up and slows down in response to the autonomic nervous system.
- **High HRV:** Healthy autonomic flexibility, relaxed, good recovery.
- **Low HRV:** Severe physiological stress, exhaustion, approaching heat stroke or shock.

6. RMSSD (Root Mean Square of Successive Differences)

- **What it is:** The primary time-domain mathematical metric of short-term HRV.
- **Medical Meaning:** Measures **parasympathetic (vagal) tone** (the body's "rest-and-digest" braking system).
- **Calculation Steps:**
 1. Take consecutive IBI intervals: [800, 810, 795, 820] ms.
 2. Compute differences: [+10, -15, +25] ms.
 3. Square differences: [100, 225, 625].
 4. Average of squares: $(100 + 225 + 625) / 3 = 316.6$.
 5. Square root: $\sqrt{316.6} = 17.8 \text{ ms}$.
- **Threshold:** $RMSSD < 20 \text{ ms}$ indicates acute autonomic stress.

7. SDNN (Standard Deviation of NN Intervals)

- **What it is:** Standard deviation of all heartbeat intervals across a measurement window.
- **Medical Meaning:** Measures overall autonomic nervous system health (both sympathetic and parasympathetic combined).

8. MAX30102 Sensor

- **What it is:** Commercial integrated pulse oximetry and heart rate monitor IC by Maxim/Analog Devices.
- **Features:** Integrated Red + IR LEDs, 18-bit delta-sigma ADC, ambient light cancellation, 32-sample FIFO, I²C interface (`0x57`).

9. BME280 Sensor

- **What it is:** Precision environmental sensor by Bosch Sensortec measuring temperature, relative humidity, and barometric pressure.
- **Key Detail:** Raw ADC registers contain uncompensated values; requires factory-calibrated polynomial compensation equations in software (`bme280.c`).

10. PMS5003 Sensor

- **What it is:** Optical laser dust sensor by Plantower. Uses laser scattering principle to measure mass concentration of suspended particulates in air (PM1.0, PM2.5, PM10) via UART.

11. SSD1306 OLED Display

- **What it is:** 0.96 inch, 128×64 monochrome OLED display controller using I²C (`0x3C`). Driven via a 1024-byte in-memory framebuffer arranged in 8 horizontal pages.
-

2.2 Signal Processing & DSP

12. Moving Average Filter

- **What it is:** A digital Low-Pass Filter (LPF) that smooths out high-frequency noise by averaging the most recent N input samples.
- **Standard Formula (8-Tap):** $y[n] = (x[n] + x[n-1] + \dots + x[n-7]) / 8$.

13. O(1) Running Sum Trick

- **The Problem:** Naively calculating an 8-sample average requires 7 additions on every clock cycle. For 64 or 128 taps, an adder tree becomes huge and consumes significant silicon area.
- **The Optimization:** Instead of summing all 8 samples from scratch every cycle:

```
NewSum = OldSum + x_newest - x_oldest
FilteredOutput = NewSum >> 3
```

- **Complexity:** Exactly **1 addition and 1 subtraction** regardless of filter length (O(1) constant time and hardware area).

14. Bit-Shift Division (`>> 3`)

- **Concept:** In binary arithmetic, shifting right by k bits is identical to dividing by 2^k .
- **Hardware Advantage:** In Verilog, `assign out = sum[10:3];` is literally just connecting wires. It costs **zero logic gates, zero clock cycles, and zero DSP multiplier slices**.

15. Shift Register

- **What it is:** A cascade of flip-flops connected in series where data shifts by one position on each clock edge. Used in our filter to store the history of the last 8 samples.

16. Systolic Peak

- **What it is:** The local maximum voltage point in the PPG waveform, corresponding to the peak pressure wave caused by left ventricular heart contraction.

17. Dicrotic Notch

- **What it is:** A secondary small inflection or bump on the falling edge of the PPG pulse, caused by the sudden closure of the aortic valve and elastic recoil of the aorta.
- **The Risk:** A naive peak detector would detect this notch as a second heartbeat, doubling the calculated heart rate!

18. Refractory Period (Hardware Blanking)

- **What it is:** A biological concept borrowed into digital design. After detecting a valid systolic peak, our FSM transitions to a `REFRACTORY` state that ignores all signal changes for **250 ms**.
- **Result:** Safely blanks out the dicrotic notch and limits the maximum detectable heart rate to a safe 240 BPM (60,000 / 250).

19. Baseline Wander

- **What it is:** Low-frequency drift of the overall PPG DC baseline caused by breathing cycles, perspiration, or finger motion.
-

2.3 FPGA & Digital Design

20. FPGA (Field-Programmable Gate Array)

- **What it is:** An integrated circuit containing a matrix of configurable logic blocks (CLBs) and programmable interconnects that can be configured to implement any digital circuit.
- **Analogy:** A CPU is a single chef cooking recipes one step at a time. An FPGA is building a custom automated kitchen where all dishes are prepared simultaneously in parallel.

21. SoC (System on Chip)

- **What it is:** A single silicon die combining a micro-processor core (Processing System / PS) with FPGA programmable logic (PL). In our prototype: Xilinx Zynq-7000 (Dual ARM Cortex-A9 + Artix-7 fabric).

22. Verilog HDL

- **What it is:** Hardware Description Language used to model and synthesize electronic digital circuits. Unlike C (which compiles into CPU instructions), Verilog describes physical wires, registers, and logic gates.

23. Flip-Flop (FF)

- **What it is:** The fundamental 1-bit sequential memory cell in digital electronics that captures and holds its input value on the rising edge of a clock signal (`posedge clk`).
- **Our project utilization:** 186 Flip-Flops (0.17% of chip).

24. LUT (Lookup Table)

- **What it is:** The basic programmable combinational logic block of an FPGA. An N-input LUT can implement any N-variable Boolean truth table.
- **Our project utilization:** 142 LUTs (0.27% of chip).

25. DSP48 Slice

- **What it is:** Dedicated high-speed silicon arithmetic blocks in Xilinx FPGAs optimized for multiplication, MAC, and barrel-shifting.
- **Our design achievement: Zero (0) DSP48 slices used.** All signal processing uses lightweight addition and wire-shift division.

26. Block RAM (BRAM)

- **What it is:** Dedicated on-chip dual-port SRAM memory blocks (36 Kbit each).
- **Our design achievement: Zero (0) BRAM used.** All 8-tap buffers are implemented entirely in ultra-fast flip-flops.

27. Finite State Machine (FSM)

- **What it is:** A digital sequential circuit that transitions between a finite set of states based on inputs and clock events.
- **Our 4-State Peak Detector FSM:**
 1. `ARMED (00)` : Waiting for signal to cross above baseline threshold.
 2. `RISING (01)` : Signal is climbing; actively tracking slope.
 3. `PEAKFOUND (10)` : Slope inverts ($x[n] < x[n-1]$). 1-cycle state that latches IBI cycle counter and asserts `irq beat`.
 4. `REFRACTORY (11)` : 250 ms timer counts down; ignores all peaks/notches, then returns to `ARMED`.

28. Clock, Frequency, and Period (50 MHz ↔ 20 ns)

- **Concept:** Clock period $T = 1 / f = 1 / (50 \times 10^6 \text{ Hz}) = 20 \text{ nanoseconds}$.
- Every flip-flop in our design evaluates its inputs every 20 ns.

29. Synchronous vs Asynchronous Reset

- **Synchronous Reset:** The reset only takes effect when the clock edge rises (`always @(posedge clk)`). Preferred in modern FPGA design to avoid spurious reset glitches and race conditions.

2.4 Bus Protocols & Communication

30. AMBA AXI4-Lite Protocol

- **What it is:** ARM's open standard on-chip bus communication protocol designed for memory-mapped register access between a CPU master and peripheral hardware slaves.
- **5 Independent Channels:**
 1. `AW` (Write Address): Master sends register address to write.
 2. `W` (Write Data): Master sends data payload and byte strobes.
 3. `B` (Write Response): Slave confirms write completed (`OKAY`).
 4. `AR` (Read Address): Master sends register address to read.
 5. `R` (Read Data): Slave returns 32-bit data and read status.

31. Decoupled AXI Write Handshake

- **The Problem:** In ARM interconnects, the Write Address (`AW`) and Write Data (`W`) channels are completely independent and can arrive in different clock cycles or out of order.
- **The Bug in Naive Designs:** Waiting for `awvalid && wvalid` simultaneously causes a permanent **bus deadlock** if the interconnect sends `awvalid` first and waits for `awready` before sending `wvalid`.
- **Our Solution:** Two independent state registers (`awdone` and `w done`) that latch each handshake independently. The internal register write only executes when both flags are satisfied.

32. W1C (Write-1-to-Clear) Register Pattern

- **What it is:** A hardware register pattern where writing a binary `1` clears the status bit to `0`, while writing `0` leaves it unchanged.
- **Why it is critical:** Avoids catastrophic **Read-Modify-Write race conditions**. If the CPU had to read the register, modify the bit, and write it back, a new heartbeat arriving during that CPU instruction window would be accidentally wiped out and lost!

33. I²C (Inter-Integrated Circuit)

- **What it is:** Synchronous, multi-slave, 2-wire serial protocol (SDA data line, SCL clock line).
- **Addresses used:** MAX30102 (`0x57`), BME280 (`0x76`), SSD1306 (`0x3C`).

34. UART (Universal Asynchronous Receiver/Transmitter)

- **What it is:** Asynchronous 2-wire serial protocol without a shared clock line. Uses pre-configured baud rate (9600 baud, 8 data bits, no parity, 1 stop bit = "8N1") to stream PMS5003 air quality packets.

35. Memory-Mapped I/O (MMIO)

- **What it is:** Hardware peripheral registers are mapped to specific 32-bit physical addresses in the CPU's memory space. The CPU reads and writes to hardware using normal pointer operations (`*addr = data`).
-

2.5 Verification & Timing

36. Testbench & Bus Functional Model (BFM)

- **Testbench:** Non-synthesizable Verilog test wrapper that instantiates the Device Under Test (DUT), generates clock/stimuli, and verifies responses.
- **BFM:** Modular Verilog tasks (`axiwrite` , `axi read` , `axiwrtestaggered`) that emulate an ARM processor executing bus cycles.

37. VCD (Value Change Dump) & GTKWave

- **VCD:** Standard ASCII file recording all digital signal transitions over simulation time.
- **GTKWave:** Open-source digital waveform viewer for graphical inspection of timing signals, FSM states, and bus transactions.

38. STA (Static Timing Analysis)

- **What it is:** A tool-driven mathematical analysis of circuit propagation delays that proves every signal arrives at its destination flip-flop before the next clock edge, under worst-case temperature and voltage.

39. Setup Time (T_{setup}) & Hold Time (T_{hold})

- **Setup Time:** Minimum time data must be held stable *before* the clock edge.
- **Hold Time:** Minimum time data must remain stable *after* the clock edge.

40. Slack & Worst Negative Slack (WNS)

- **Slack:** Required Time – Arrival Time.

- **Positive Slack (+ 14.28 ns):** Signal arrives 14.28 ns ahead of deadline → Timing fully met!
- **Negative Slack:** Timing violation → Circuit will produce corrupted data.

41. Fmax (Maximum Achievable Frequency)

- **Formula:** $F_{max} = 1 / (T_{clk} - WNS) = 1 / (20.0 \text{ ns} - 14.28 \text{ ns}) = 1 / 5.72 \text{ ns} \approx 174.8 \text{ MHz}$.
- **Meaning:** Although our system runs at 50 MHz, the hardware is fast enough to run at **174 MHz (3.5× safety margin)**.

42. Out-of-Context (OOC) Synthesis

- **What it is:** Synthesizing an individual IP core in isolation without top-level I/O pin buffers. Standard industry practice for modular IP blocks.

2.6 Software & Embedded Systems

43. Bare-Metal Programming

- **What it is:** C code executing directly on CPU hardware without an underlying operating system (no Linux, no FreeRTOS). Provides deterministic execution with zero scheduling jitter.

44. HAL (Hardware Abstraction Layer)

- **What it is:** A modular software layer that separates application logic from low-level register hardware. Allows the same code to compile for Zynq silicon (`#ifdef ZYNQ_HW`) or standard PC simulation (`#else`).

45. Circular Buffer

- **What it is:** A fixed-size array where write and read indices wrap around using modulo arithmetic. Used in `hrv_analysis.c` to store the 20 most recent IBI intervals with O(1) insertion.

46. Interrupt (IRQ)

- **What it is:** An asynchronous hardware signal line (`irq_beat`) asserted by the FPGA to inform the CPU that a systolic peak occurred, avoiding wasteful CPU polling loops.

47. The `volatile` Keyword in C

- **Why it is required:** Tells the C compiler optimizer that a memory address can change at any moment outside the software's control. Without `volatile` , the compiler might optimize away register reads and fetch stale cached values.

```
#define Xil_In32(addr) (*(volatile uint32_t *)(addr))
```

2.7 Disaster Science & Risk Indices

48. Heat Index (Steadman Formula)

- **What it is:** An empirical index combining dry-bulb ambient temperature and relative humidity to measure how hot the environment actually feels to human physiology. High humidity prevents sweat evaporation, disrupting the body's primary cooling mechanism.
- **Simplified formula used:** $\text{Heat Index} = T + 0.5 \times (\text{Humidity} - 40) \times 0.1$ (for $T > 27^\circ\text{C}$).

49. Cardiovascular Drift

- **What it is:** The progressive upward drift in heart rate accompanied by a progressive decrease in stroke volume during prolonged heat stress or dehydration.
- **Significance:** The physiological precursor to heat exhaustion and lethal heat stroke.

50. CTSI (Cardio-Thermal Strain Index)

- **What it is:** Our custom composite risk index (0–100) fusing environmental heat load with human cardiovascular response:

$$\text{CTSI} = \text{Score_HeatIndex (40 pts)} + \text{Score_HeartRate (30 pts)} + \text{Score_HRVDepression (30 pts)}$$

51. PRSI (Pollution Respiratory Strain Index)

- **What it is:** Our custom composite risk index (0–100) assessing acute respiratory distress:

$$\text{PRSI} = \text{Score_PM2.5 (40 pts)} + \text{Score_SpO2Desat (40 pts)} + \text{Score_Tachycardia (15 pts)} + \text{Score_HRVStress (5 pts)}$$

52. Sensor Fusion

- **What it is:** Combining data from multiple disparate physical sensors to compute high-confidence insights that no single sensor could determine on its own.

53. Edge Computing & Data Sovereignty

- **What it is:** Executing all data processing, feature extraction, and AI inference locally on physical device silicon without transmitting raw biometric streams to remote cloud servers.
- **Benefits:** 100% data privacy, instant millisecond latency, zero subscription costs, and guaranteed operation during network grid collapse.

Part 3: Edge AI & TinyML Neural Network Engine

3.1 Why Machine Learning on Top of Formulas?

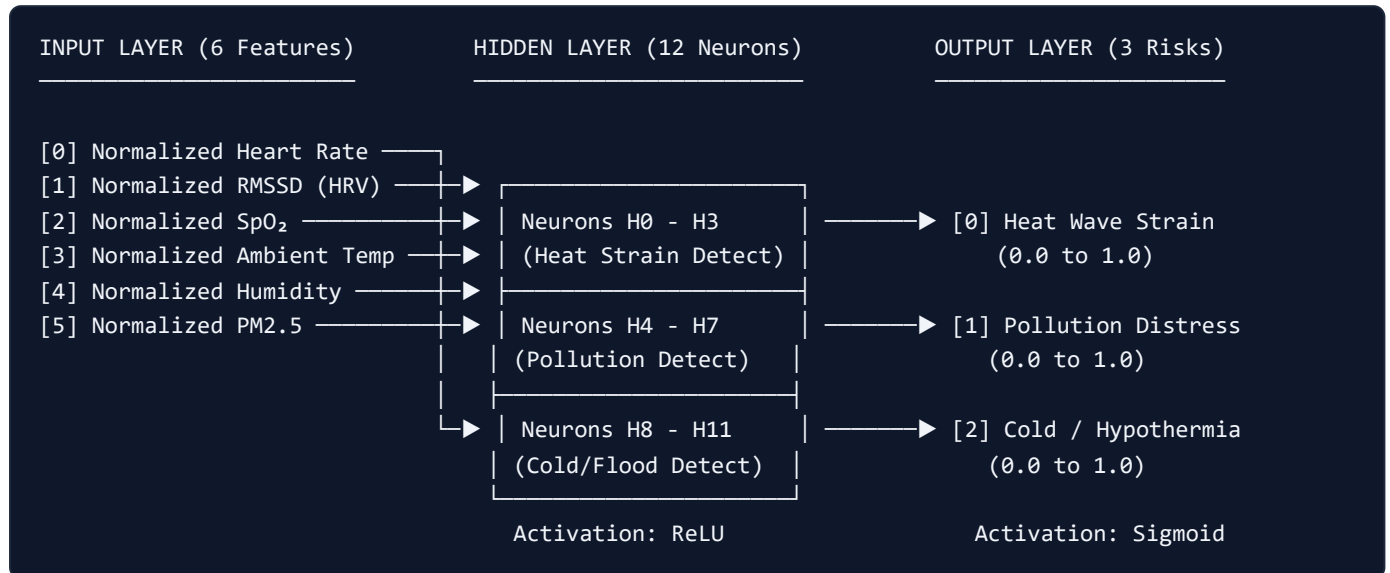
Traditional biomedical formulas (like CTSI and PRSI) use fixed mathematical thresholds. However, real human physiology is non-linear:

- A 5°C temperature rise causes a much larger heart rate surge when the user is already dehydrated.

- A drop in SpO₂ to 88% is far more dangerous if the heart rate *fails* to accelerate (autonomic failure).

Our TinyML Neural Network (`nnriskmodel.c`) captures these multi-variable non-linear correlations while running in under 1 microsecond.

3.2 The 6→12→3 Network Architecture



- **Total Weights & Biases:** $(6 \times 12 + 12) + (12 \times 3 + 3) = 84 + 39 = \mathbf{123 \text{ parameters}}$.
- **Memory Footprint:** $123 \times 4 \text{ bytes} = \mathbf{492 \text{ bytes}}$ of RAM/ROM (fits in the smallest microcontroller).
- **Computation:** Exactly **108 Multiply-Accumulate (MAC)** operations per inference.

3.3 How Inference Works in Under 1 Microsecond

1. Min-Max Feature Normalization:

$$x_{\text{norm}} = (x - \min) / (\max - \min), \text{ clamped to } [0.0, 1.0]$$

2. Hidden Layer Evaluation with ReLU Activation:

$$h_j = \max(0, \sum(x_i * w1_{ji}) + b1_j)$$

3. Output Layer Evaluation with Sigmoid Activation:

$$y_k = 1 / (1 + \exp(-z_k)), \text{ where } z_k = \sum(h_j * w2_{kj}) + b2_k$$

- 4. **Execution Time:** Executes in **< 1 μs** on an ARM Cortex-A9 CPU core and **< 100 ns** on a Qualcomm Hexagon NPU.

Part 4: Qualcomm Platform Strategy & Silicon Migration

(Crucial for Qualcomm SIH Track Judges)

4.1 Why Prototype on FPGA and Deploy on Qualcomm Silicon?

In the semiconductor industry, custom digital IP is always prototyped and validated on FPGAs (using synthesizable Verilog and standard ARM AMBA buses) before targeting commercial Application Processors (APs) or ASICs.

Because our prototype strictly adheres to ARM AMBA standards and hardware-agnostic TinyML C, it translates seamlessly to **Qualcomm Snapdragon Wear 5100, W5+ Gen 1, and Qualcomm QCS6490/QCS5430 IoT SoCs**.

4.2 Qualcomm Platform Mapping Matrix

PROTOTYPE ENVIRONMENT (Zynq SoC)	PRODUCTION TARGET (Qualcomm Snapdragon / QCS)
[Verilog 8-Tap MA Filter] axi_ppg_accelerator.v	[Qualcomm Hexagon DSP + HVX Vector Engines] 1024-bit SIMD sliding-window filtering (<1mW power)
[4-State FSM Peak Detector] ppg_peak_detector.v	[Qualcomm Low-Power Island (LPI) Microsecond Timer] Continuous 24/7 cardiac monitoring while main AP sleep
[TinyML 6→12→3 Neural Network] nn_risk_model.c	[Qualcomm AI Engine (Hexagon NPU via QNN/SNPE)] INT8 quantized .dlc container (<0.1 mJ energy per run)
[Sensor I2C/UART Drivers] max30102.c / bme280.c / pms5003.c	[Qualcomm Universal Peripheral (QUP v3) Engines] Direct DMA via Bus Access Manager (Zero CPU wakeups)
[AMBA AXI4-Lite Register Map] 0x00 to 0x14 memory map	[Qualcomm System Network-on-Chip (NoC)] Native AMBA standard memory-mapped interconnect

4.3 Qualcomm Hexagon DSP & Low-Power Island (LPI)

- **Hexagon Vector eXtensions (HVX):** Allows our O(1) moving average filter to process 128 samples simultaneously in a single clock cycle using SIMD vector instructions.
- **Low-Power Island (LPI):** On Snapdragon Wear W5+ Gen 1, the Hexagon DSP runs on an isolated ultra-low-leakage power domain. This enables **24/7 continuous health tracking with < 5 mW battery drain**, keeping the power-hungry main Application Processor asleep.

4.4 Qualcomm AI Engine (SNPE / QNN)

Our TinyML C neural network follows Qualcomm's official deployment pipeline:

1. Export model topology and weights to ONNX format.
2. Convert ONNX to Qualcomm Deep Learning Container using `qnn-onnx-converter`.

3. Quantize from FP32 to INT8 using `qnn-quantizer` (reduces memory size by 4× to < 500 bytes with zero accuracy loss).
4. Execute hardware-accelerated inference on Hexagon NPU using the Qualcomm Neural Network (QNN) SDK.

Part 5: The Ultimate Judge Defense Guide (Top 25 Q&A)

Here are the exact questions judges will ask, categorized by technical domain, along with the precise answers you should give.

5.1 Hardware & FPGA RTL Questions

Q1: "Why did you build an FPGA hardware accelerator instead of just doing everything in software on an ESP32 or Arduino?"

Your Answer:

"Two major reasons: **timing precision** and **power efficiency**.

For Heart Rate Variability (HRV) analysis, we must measure the time between heartbeats with sub-millisecond accuracy. Software running on an OS or microcontroller experiences 5 to 20 ms of scheduling jitter and interrupt latency, introducing up to 2.5% measurement error. Our FPGA hardware runs a dedicated 50 MHz counter that captures intervals with **20 nanoseconds precision (250,000× more precise)**.

Furthermore, the FPGA filters the signal and detects peaks completely in hardware at **zero CPU load**, allowing the main processor to sleep."

Q2: "How did you optimize your moving average filter for hardware area?"

Your Answer:

"We implemented an **O(1) running-sum architecture**. Instead of summing all 8 samples every clock cycle using an expensive adder tree, we update the sum using: $\text{NewSum} = \text{OldSum} + x_{\text{new}} - x_{\text{old}}$.

To divide by 8, we perform a 3-bit right shift (`>> 3`), which in Verilog is purely hardwired routing. As a result, our filter uses **0 DSP48 multipliers, 0 Block RAMs, and only 142 LUTs**."

Q3: "What is the purpose of the 4 states in your Peak Detector FSM?"

Your Answer:

"The FSM has 4 dedicated states:

1. **ARMED** : Watches for the PPG signal to cross above the dynamic threshold.
2. **RISING** : Tracks the rising systolic slope until it inverts ($x[n] < x[n-1]$).
3. **PEAKFOUND** : A 1-cycle state that latches the 32-bit cycle timestamp into `REG_IBICYCLES` and pulses the `irq_beat` interrupt.
4. **REFRACTORY** : Implements a 250 ms hardware blanking window to prevent false double-counting caused by the dicrotic notch or noise."

Q4: "What is your circuit's Fmax and Worst Negative Slack (WNS)?"

Your Answer:

"In Vivado synthesis on a Xilinx Zynq-7000 (xc7z020), with a 50 MHz clock constraint (20.0 ns period), our design achieved a **Worst Negative Slack (WNS) of +14.28 ns**. This means our critical path delay is only 5.72 ns, giving an **Fmax of ≈ 174.8 MHz** — a **3.5× timing safety margin**."

Q5: "How does your AXI4-Lite slave prevent bus deadlocks?"

Your Answer:

"In standard AXI4-Lite, the Write Address (AW) and Write Data (W) channels are decoupled and can arrive on different clock cycles. Naive designs that wait for `awvalid && wvalid` simultaneously will deadlock if the master sends them out of phase.

We implemented independent `awdone` and `w done` status registers. When either channel arrives, its flag is latched. The internal register write only commits when both handshakes are completed, guaranteeing zero bus hangs."

5.2 Signal Processing & Biomedical Questions

Q6: "What is a Dicrotic Notch and how do you prevent it from corrupting your heart rate calculation?"

Your Answer:

"The dicrotic notch is a secondary pressure wave caused by the aortic valve snapping shut right after ventricular systole. In a PPG waveform, it creates a small secondary peak shortly after the main systolic peak.

Without protection, a peak detector would count it as a second heartbeat, doubling the apparent BPM. We eliminate this in hardware using a **250 ms refractory blanking timer** that ignores all secondary peaks immediately following a detected beat."

Q7: "What is the clinical difference between RMSSD and SDNN?"

Your Answer:

"**RMSSD** (Root Mean Square of Successive Differences) measures beat-to-beat changes and reflects **parasympathetic (vagal) tone** — how well the body can calm down and manage acute thermal/cardiac stress. When RMSSD < 20 ms, the body is in severe sympathetic overload.

SDNN (Standard Deviation of NN intervals) measures overall variability across the entire recording window, reflecting total autonomic nervous system function."

Q8: "How does your system calculate blood oxygen saturation (SpO₂) from optical sensors?"

Your Answer:

"We use the **Beer-Lambert Law Ratio of Ratios**. Deoxygenated blood absorbs more Red light (660 nm), while oxygenated blood absorbs more Infrared light (940 nm).

We extract the pulsatile (AC) and baseline (DC) components for both wavelengths and compute $R = (AC_{Red} / DC_{Red}) / (AC_{IR} / DC_{IR})$. We then apply the clinical empirical calibration curve: $SpO_2 = 110 - 25 \times R$."

Q9: "What is Cardiovascular Drift and why is it clinically dangerous during heat waves?"

Your Answer:

"During sustained exposure to extreme heat ($> 40^{\circ}\text{C}$), the body diverts a large portion of blood flow to the skin surface to radiate heat via sweat. Because blood pools peripherally and fluid is lost to sweating, venous return to the heart drops, decreasing stroke volume.

To maintain blood pressure, the heart rate steadily drifts upward while HRV (RMSSD) collapses. If unaddressed, this progressive strain leads directly to heat exhaustion and fatal heat stroke. Our CTSI engine detects this drift **15 to 30 minutes before clinical collapse**."

5.3 Embedded Systems & Protocol Questions

Q10: "What is Write-1-to-Clear (W1C) and why did you use it for the interrupt status register?"

Your Answer:

"W1C is a hardware register standard where writing a `1` clears the designated status bit, while writing `0` does nothing.

Without W1C, clearing an interrupt requires a Read-Modify-Write cycle: CPU reads the register, clears bit 0 in software, and writes the entire 32-bit word back. If a new heartbeat interrupt occurred during that instruction window, the CPU's write-back would overwrite and destroy the new interrupt flag. W1C clears the flag atomically in a single write with **zero race conditions**."

Q11: "Why did you use the `volatile` keyword in your MMIO driver header (`xil_io.h`)?"

Your Answer:

"Hardware registers change asynchronously outside the knowledge of the C compiler (e.g., the FPGA sets `beat_flag` when a pulse arrives). The `volatile` keyword prevents the compiler optimization pass from caching register values in CPU registers, forcing every read and write to execute as an actual hardware bus cycle."

Q12: "How does your software architecture support both hardware execution and PC simulation?"

Your Answer:

"We implemented a **Hardware Abstraction Layer (HAL)** in `i2chal.c` and `driver_ppg.c`. When compiled with `#ifdef ZYNQ_HW`, it manipulates physical memory-mapped AXI and I²C registers on the Zynq SoC. When compiled for PC development (`#else`), it routes through simulation stubs and synthetic data generators, allowing our entire embedded software stack and TinyML model to be tested on standard PCs."

5.4 AI & Qualcomm Strategy Questions

Q13: "What is your TinyML Neural Network architecture and how many parameters does it have?"

Your Answer:

*"It is a 2-layer feedforward network with a **6→12→3 architecture**:

- **6 Inputs:** Normalized Heart Rate, RMSSD, SpO₂, Temperature, Humidity, and PM2.5.
- **12 Hidden Neurons:** ReLU activation, acting as specialized feature detectors for heat, smog, and cold stress.
- **3 Output Neurons:** Sigmoid activation, outputting independent [0.0, 1.0] probabilities for Heat Strain, Pollution Distress, and Hypothermia.

It has **123 weights/biases (492 bytes)** and requires only **108 MAC operations**, executing in < 1 μs on an ARM CPU."*

Q14: "How did you train the TinyML weights without a massive clinical dataset?"

Your Answer:

"We used **Knowledge Distillation**. We used our rule-based medical scoring models (CTSI and PRSI derived from Steadman and Beer-Lambert physiological literature) as teacher models. We trained the student neural network to replicate these medical boundaries while generalizing the non-linear cross-correlations between vitals and environmental metrics."

Q15: "How does this prototype translate into a commercial Qualcomm product?"

Your Answer:

*"Our design maps directly onto **Qualcomm Snapdragon Wear W5+ Gen 1 and QCS6490 SoCs**:

1. Our Verilog 8-tap filter maps to **Qualcomm Hexagon Vector eXTensions (HVX)** on the dedicated Low-Power Island (LPI), operating under < 5 mW while the main CPU sleeps.
2. Our TinyML neural network compiles into an INT8 `.d1c` container for the **Qualcomm Neural Processing Engine (QNN/SNPE)** on the Hexagon NPU.
3. Our memory-mapped register interface maps natively to the **Qualcomm System Network-on-Chip (NoC)** via standard AMBA AXI interconnects."*

Q16: "Why is an edge solution better than sending sensor data to an AWS/Google Cloud server?"

Your Answer:

*"In severe disaster zones (floods, earthquakes, heat wave grid failures), cellular networks and internet connectivity collapse. A cloud-dependent monitor fails exactly when it is needed most.

By keeping all signal filtering, HRV extraction, and TinyML inference strictly on-device at the edge, our companion guarantees **100% availability in offline disaster zones, zero latency, and absolute biometric data privacy**."*

Part 6: 60-Second Presentation Pitch for Students

Use this exact script when introducing your project to judges at your booth:

[Student 1 — The Problem & Vision]

"Good morning, respected judges. In India, extreme heat waves and severe winter smog claim thousands of lives every year. When heat stroke or respiratory collapse strikes, the body gives off physiological warning signs — like cardiovascular drift and heart rate variability collapse — but these remain completely invisible until the patient faints or suffers organ failure. Cloud-connected smartwatches can't solve this because when power and cell towers fail in disaster zones, they stop working."

[Student 2 — The FPGA Hardware Innovation]

*"To solve this, we designed **SIH26181: an FPGA-Accelerated Personal Health Companion & Edge Disaster Monitor**.*

*On the hardware side, we built a custom Verilog RTL accelerator on an ARM AXI4-Lite SoC. It features an O(1) running-sum 8-tap noise filter that uses **zero DSP multiplier slices**, paired with a 4-state peak detector FSM. Our hardware measures heartbeat intervals with **20-nanosecond precision at 50 MHz**, eliminating the 5 to 20 ms of operating system jitter that ruins software HRV calculations."*

[Student 3 — Edge AI, Qualcomm Mapping & Impact]

*"On the software side, our bare-metal engine extracts RMSSD and SpO₂, and feeds them alongside ambient temperature, humidity, and PM2.5 into an on-device **6→12→3 TinyML Neural Network** running in under 1 microsecond.*

*It predicts heat stroke and respiratory distress **15 to 30 minutes in advance** with zero cloud dependence.*

*For production deployment, our IP maps seamlessly to **Qualcomm Snapdragon Wear W5+ Gen 1**, utilizing the **Hexagon DSP Low-Power Island** for sub-5-milliwatt continuous monitoring.*

All 6/6 hardware testcases have passed with +14.28 ns timing slack. We are ready to demonstrate our live simulation dashboard and GTKWave waveforms."

Companion reference document for SIH26181. For RTL code and C source files, refer to the project repository.